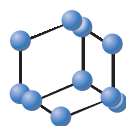


REVIEW ARTICLE

BENTHAM
SCIENCE

Antibiotics and Antibiotic Resistance- Flipsides of the Same Coin

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Abstract: One of the major global health care crises in the 21st century is antibiotic resistance. Almost all clinically used antibiotics have resistance emerging to them. Antibiotic Resistance can be regarded as the 'Faceless Pandemic' that has enthralled the entire world. It has become peremptory to develop treatment options as an alternative to antibiotic therapy for combating antibiotic-resistant pathogens. A clearer understanding of antibiotic resistance is required to prevent the rapid spread of antibiotic-resistant genes and the re-emergence of infections. The present review provides an insight into the different classifications and modes of action of antibiotics to understand how the hosts develop resistance to them. In addition, the association of genetics in the development of antibiotic resistance and environmental factors has also been discussed, emphasizing developing action plans to counter this "quiescent pandemic". It is also pertinent to create models that can predict the early resistance so that treatment strategies may build up in advance with the evolving resistance.

Keywords: Antibiotic resistance, *de novo* mutations, antibiotic stewardship, quiescent pandemic, environmental factors, modes of action.

1. INTRODUCTION

Antibiotic resistance (ABR) is a natural phenomenon where microorganisms become resistant to antibiotics and threaten the prevention and treatment of several diseases caused due to a range of microorganisms [1]. The process of resistance is natural, but it is being driven by the exploitation of antibiotics in the environmental, healthcare, and agriculture fields. The phenomenon of ABR is very primitive. There is evidence from the permafrost samples of Yukon suggesting the presence of resistant mutations in bacteria 30,000 years before the discovery of the first-ever antibiotic, Penicillin [2]. The organisms develop and evolve specific mechanisms that enable them to survive under certain concentrations of these antimicrobial compounds.

Are humans the reason behind ABR? The answer formulated by scientists over the past years of intensive research is both "yes and no." ABR is emerging so rapidly because of its primitive nature [3]. It has always been there. Microbes producing antibiotics possess immunity genes. Such orthologous genes have been identified on mobile genetic elements in resistant pathogens in clinical settings [4]. However, it is being driven due to overexploitation by humans in various fields.

ABR has made it a strenuous task to eradicate some infections and has even led to the re-emergence of other infections. The development of all antibiotics to date has created complacency regarding ABR. To eliminate the infections and develop treatment

options for various diseases, it is necessary to discover new drugs [5]. Scientists worldwide focus on multiple strategies to create non-antibiotic treatment options to combat resistance [6]. This review intends to discuss the current knowledge about the classification, mode of action of antibiotics, and genetics behind the development of antibiotic resistance. Additionally, this review also discusses the mechanism of antibiotic resistance, the emergence and re-emergence of antibiotics resistance, and factors and complications associated with it. It also discusses the various strategies being developed and successful outcomes along with the genetics and consequences, prevention, and control of ABR.

2. CLASSIFICATION OF ANTIBIOTICS

There are two approaches for detecting new antibiotics- the search for natural compounds that act as antibiotics and a parallel approach is the testing of synthetic chemicals. Synthetic antibiotics that are currently in use fall under three major classes: Sulfonamides, the first artificial antibiotic class; fluoroquinolones, the second synthetic antibiotic class; and oxazolidinones, the third class synthetic antibiotics derived *de novo* from the molecular framework, along with first-generation antibiotic linezolid [7, 8]. Microbes produce natural antibiotics in their microenvironment. Historically, the approach followed for the detection of natural antibiotics was an assay where soil samples were extracted using organic solvents out of the aqueous phase followed by fractionation procedures like Minimum inhibitory concentration (MIC) determination. Figs. (1 and 2) depict the structures of synthetic and natural antibiotics [9]. The natural antibiotics used in clinical treatments include beta-lactams, aminoglycosides, glycopeptides, tetracyclines, macrolides, polyketides, and lipopeptides. These successive generations of antibiotics were achieved starting from the 1940s [10].

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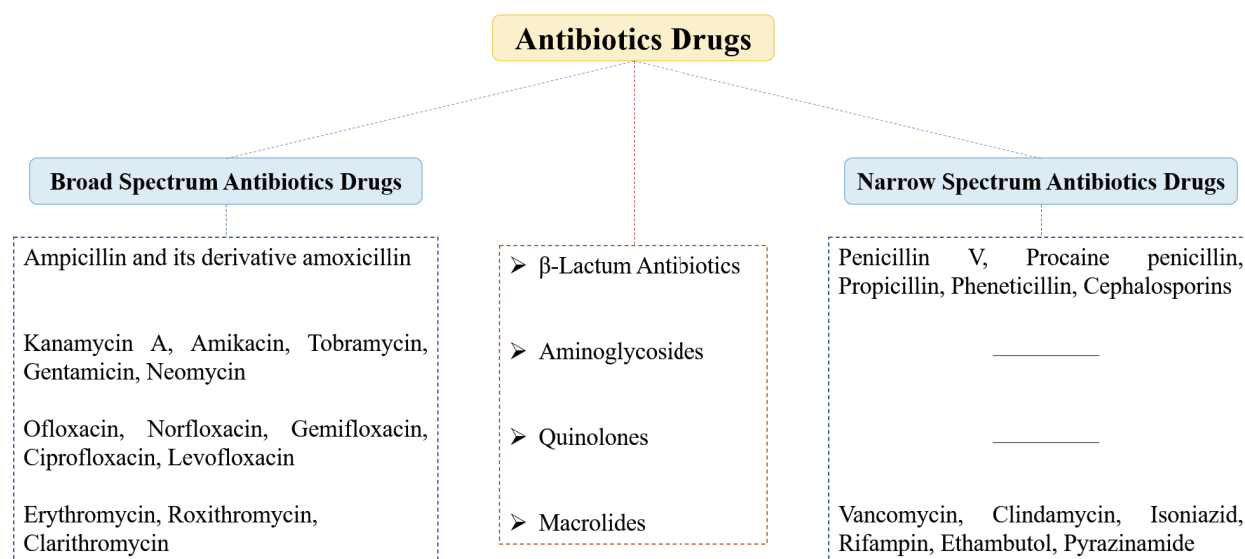


Fig. (1). Classification of antibiotics based on the spectrum of activity. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

Infectious diseases are the major causes of human sickness and death. To overcome such health care issues, antibiotics have proved to be promising agents since they were introduced in the 1940s [11]. Antibacterials, a subclass of antibiotics, have been classified earlier in several ways; however, to make it more easily understandable, we can classify antibacterial agents into five groups: type of action, source, the spectrum of activity, chemical structure, and function [12].

2.1. Classification Based on the Type of Action

Commonly, antibacterial agents are classified according to their mode of action, *i.e.*, bactericidal and bacteriostatic. In general, antibacterial agents which disrupt the cell membrane or cell wall of bacteria are stated as bactericidal. In contrast, the antibacterial agents that inhibit or slow the proliferation of bacteria are stated as bacteriostatic agents [13, 14]. The inhibition *via* bacteriostatic agents involves inhibiting bacterial metabolic pathways and protein synthesis. Since bacteriostatic agents curb the proliferation of pathogenic bacteria, sometimes it gets difficult to differentiate between bacteriocidal and bacteriostatic, primarily when bacteriostatic agents are used in high concentrations [15, 16].

2.2. Classification Based on the Source of Antibacterial Agents

Antibacterials are another subclass of antibiotics, naturally obtained from fungal and bacterial species and involve semisynthetic agents that are either synthetic or chemically altered natural products [17]. Few recognized natural antibacterials are benzylpenicillin, cefamycins, cephalosporins, and gentamicin. It has been claimed that natural antibacterials often show high toxicity compared to synthetic antibacterial agents [18]. Amikacin and ampicillin are semisynthetic antibacterial agents that exhibit increased effectiveness and low toxicity. Nowadays, promising artificial antibacterial agents are moxifloxacin and norfloxacin [11].

2.3. Classification Based on the Spectrum of Activity

The antibacterial agents are also classified based on target specificity, in which antibacterial agents may have either broad or a narrow spectrum. The terms general and narrow-spectrum apply to both the industrial and academic sectors [19]. The narrow-spectrum antibacterials are the ones that have an antibacterial effect against a limited number of microorganisms (*i.e.*, either gram-positive or gram-negative bacteria). Unlike narrow-spectrum anti-

bacterial, broad-spectrum antibacterial shows an antibacterial effect on many pathogenic bacteria of both gram-positive and gram-negative bacteria [20]. Generally, narrow-spectrum antibacterials are ideal and favored over broad-spectrum antibacterials because they kill specific microorganisms in the body and have fewer chances to cause superinfection [21]. Furthermore, narrow-spectrum antibacterials have less contribution in developing resistance in bacteria as they deal with a limited number of bacteria, as shown in Fig. (1).

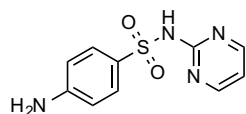
2.4. Classification Based on Chemical Structure

Different antibiotics show diverse therapeutic behavior due to their chemical differences, which has made it ultimately essential to classify the antibacterial agents according to their chemical structure [22]. This classification is very vital as a similar structural agent exhibits similar toxicity pattern, effectiveness, and other associated properties. Generally, based on structure, antibacterial agents are classified into two groups, *i.e.*, Group A involving β -lactams and Group B involving aminoglycosides [23]. In a more elaborative way, antibacterial agents can be classified as aminoglycoside, β -lactams, β -lactam/ β -lactamase inhibitor combinations, fluoroquinolones, macrolides, and quinolones.

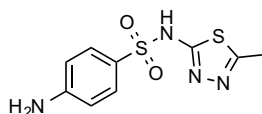
Sulphonamides are a class of important synthetic organic compounds having a sulphonamide functional group ($R_1-SO_2-NR_2R_3$) in their structures and are known to have medical importance [24]. Few compounds of this group are known to exhibit antibacterial activity like sulfadiazine. Thiazide diuretics and sulfonylureas are newly proposed drug groups based on antibacterial sulphonamides, as depicted in Fig. (2).

Aminoglycosides are compounds composed of two amino sugars linked to an aminocyclitol *via* a glycosidic bond [25]. Amikacin, gentamicin, kanamycin, neomycin, netilmicin, paromomycin, sisomicin, spectinomycin, streptomycin, tobramycin, and soframycin are commonly used aminoglycosides, as shown in Fig. (2) [26].

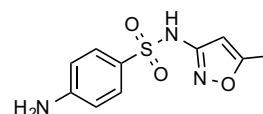
Beta-lactams (β -lactams) are the most popular drug class, composed of four-membered lactam rings, which vary based on attached additional cycles or side chains [27, 28]. Carbapenems, cephalosporins, imipenem, monobactams, and penicillin derivatives are a few examples of Beta-lactam drugs illustrated below in Fig. (2) [29].

Sulphonamides-

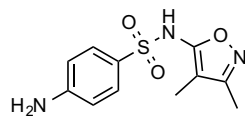
Sulfadiazine



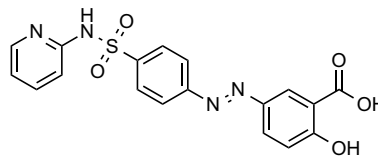
Sulfamethizole



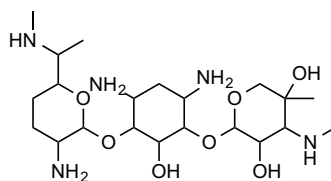
Sulfamethoxazole



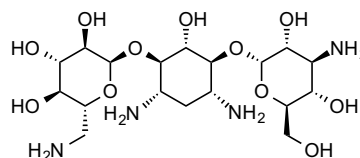
Sulfisoxazole



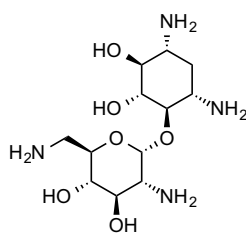
Sulfasalazine

Aminoglycosides-

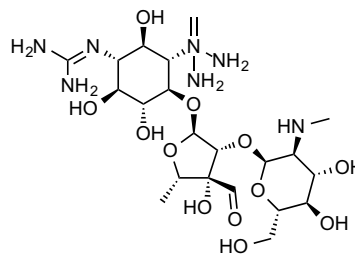
Gentamicin



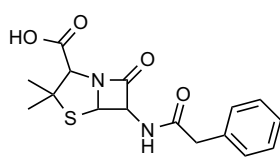
Kanamycin



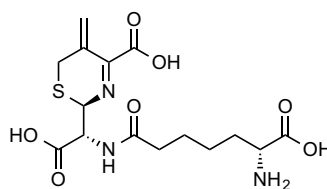
Neomycin



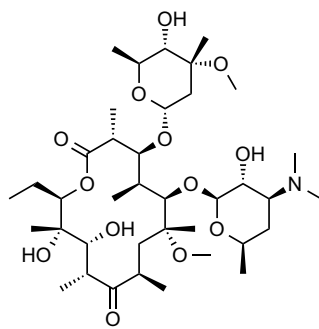
Streptomycin

 β -Lactams-

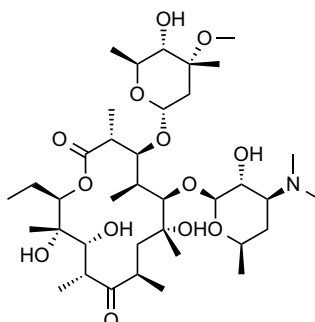
Penicillin



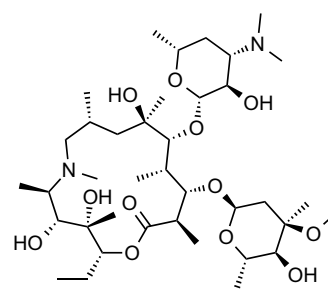
Cephalosporin

Macrolide-

Clarithromycin



Erythromycin



Azithromycin

(Fig. 2) Contd....

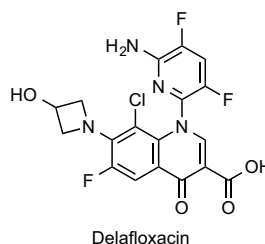
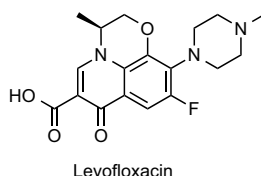
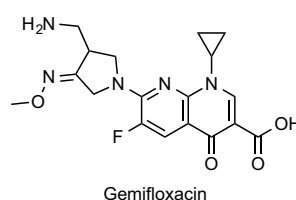
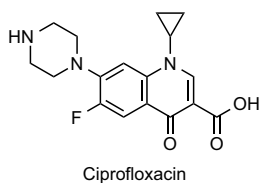
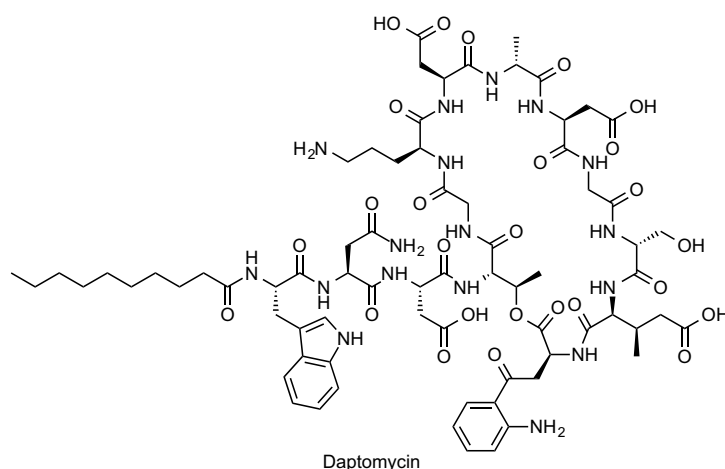
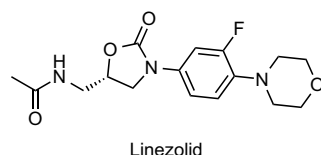
Quinolones and Fluoroquinolones-**Oxazolidinone-**

Fig. (2). Classification of antibiotics based on chemical structure.

Macrolides are a member of the class polyketide of natural products composed of macrocyclic lactone rings ranging from 14-16 to which one or more deoxy sugars, predominantly desosamine and cladinose, are attached [15, 30]. Erythromycin and roxithromycin are well-known and extensively used macrolides, as described in Fig. (2).

Quinolones are another class of antibacterial agents composed of quinine-derived structural units. The quinolone fluorine at position six is stated as fluoroquinolone. Interestingly, the variation at positions 1-, 5-, 6-, 7-, and 8- in the bicyclic rings alters the therapeutic behavior of these drugs [31]. Generally, alteration in the structure of these quinolones led to improved potency, pharmacokinetics, and coverage of antibacterial agents like garenoxacin and moxifloxacin [32]. Nalidixic acid (first generation), ciprofloxacin (second generation), levofloxacin (third generation), and trovafloxacin (fourth generation) are well-known quinolone drugs, as shown in Fig. (2). Moreover, the modifications in the structure of these drugs are related to various adverse effects [33].

Oxazolidinones, another new class of synthetic antibacterial agents, comprises a heterocyclic organic compound encompassing oxygen and nitrogen in a 5-membered ring. This class of drugs has

a unique mechanism of inhibiting the protein synthesis in bacteria [34]. This antibacterial type involves daptomycin and nitroimidazole [35]. The first clinically approved oxazolidinone is linezolid, which showed a practical bacteriostatic effect against numerous resistant pathogens like vancomycin-resistant *enterococci*, methicillin-resistant *Staph aureus*, and penicillin-resistant *Streptococcus pneumoniae*, as depicted in Fig. (2) [36].

Although, classification of antibiotics is being done based on their chemical structure, source, and spectrum of the drug. However, the mode of action of antibiotics is another classification of antibiotics that needs to be understood for progressive assessment of these drugs against the resistant bacteria strain.

3. MECHANISMS OF ACTION OF ANTIBIOTICS

Antibiotics are considered 'miracle drugs' [37] that have been in use for the past 85 years to treat various infections. Antibiotics have been used for almost more than a century, including the 'Salvarsan' or 'Compound 606' drug of Paul Ehrlich [38]. Antibiotics kill bacteria by different modes of action that have a potentially lethal effect on bacterial cells [39]. There are five main modes of action of an antibiotic:

3.1. Inhibition of Cell Wall Biosynthesis

Bacterial cells have an outer layer of peptidoglycan that protects them under unfavorable conditions. It is composed of cross-linked peptides of two amino sugars, NAM (N-acetylmuramic acid) and NAG (N-acetylglucosamine) and a pentapeptide [40, 41]. Transglycosylases and transpeptidases are the two enzymes that play a crucial role in synthesizing peptidoglycan through PBPs (Penicillin-binding proteins) that catalyze peptide bond formation. These enzymes add disaccharide pentapeptides to extend the glycan strands and cross-link the chains. Antibiotics like penicillin, cephalosporin, and carbapenems block the cross-linking of peptidoglycan units by inhibiting the activity of PBPs. Vancomycin inhibits peptidoglycan synthesis by blocking the action of these enzymes and binding itself to peptidoglycan units [42, 43].

3.2. Interference with Protein Synthesis

Proteins determine the structural composition, physiological processes, metabolic pathways, and responses to adverse conditions in living cells. Bacteria have two ribosomal units, the 30S (smaller subunit) and 50S (larger subunit). 5S, 16S, and 23S genes reside on the bacterial rRNA [44]. The 16S gene is present on the smaller subunit, and 5S and 23S genes are present on the larger subunit. Two drugs inhibit protein synthesis – 30S inhibitors and 50S inhibitors [45]. The 50S inhibitors block the initiation or elongation phase of protein synthesis, including clindamycin, lincomycin, erythromycin, linezolid, and chloramphenicol. Oxazolidinones interfere with the placement of aminoacyl-tRNA by interfering with the A site of the ribosome. 30S inhibitors including spectinomycin, tetracycline, and streptomycin block the access of aminoacyl-tRNA to the ribosome [46, 47].

3.3. Interference with Nucleic Acid Synthesis

Interference with the nucleic acid synthesis or function of a bacterial cell is pernicious for it. Antibiotics inhibit nucleic acid synthesis by blocking replication through interference with the double helix structure of DNA *via* helicases [48, 49]. Quinolones interfere with type II topoisomerase, DNA gyrase, and type IV topoisomerase, causing the double strand of DNA to break. Some antibiotics also inhibit the process of transcription. Rifampicin interrupts the DNA-directed RNA polymerase [49].

3.4. Disruption of Cell Membrane

Antibiotics like polymyxins interfere with bacterial permeability by binding to the lipid moiety of lipopolysaccharide in the cell. This leads to the leakage of bacterial contents. Daptomycin works by binding to the cytoplasmic membrane in a calcium-dependent manner and oligomerizing itself in the membrane. This leads to an efflux of potassium from the cell, causing inhibition of macromolecular synthesis [50, 51].

3.5. Interference with the Metabolic Pathways

Folate synthesis is crucial for the biosynthesis of nucleotides [52, 53]. Sulfonamides and trimethoprim interfere with folate synthesis by mimicking tetrahydrofolate required for cellular metabolism [54, 55].

3.6. Inhibitors of Membrane Function

The cytoplasmic membranes covering the cytoplasm help in maintaining the integrity of the cell and act as a barrier. And any hindrance in the functionality of the cytoplasmic membrane results in the elution of ions and macromolecules, which leads to cell death [56]. Thus, the selection of chemotherapy agents is crucial as they target bacterial cell membranes. For instance, polymyxins, active antibacterial agents composed of cyclic peptides with a long hydrophobic tail, are found in diverse forms ranging from A-E, where B and E are more extensively used for therapeutic purposes. The literature reveals that polymyxins exhibit their specificity against polysaccharides, predominantly found in most gram-negative

bacteria's outer membrane. Hence, it is proclaimed that polymyxins have selective toxicity against gram-negative bacteria. The mechanism of action of polymyxins involves the interaction of polymyxins with lipopolysaccharide of gram-negative bacteria outer membrane, which alters the structure of the membrane and increases the permeability, which as a resultant, causes osmotic imbalance [57]. Moreover, alterations like increased water uptake, inhibition of respiration, and liberation of cellular components from the cell's interior often lead to cell senescence. Whereas gram-positive bacteria have thick cell walls that prohibit the interaction of these antibacterial agents with the cell membrane of gram-positive bacteria, this explains why polymyxins have no or more negligible effect on gram-positive bacteria, as shown in Fig. (3) [58].

Even though the mode of action of different antibiotics is effective and unique to regulate the proliferation of pathogenic microbes, the extensive use of these antibiotics has led to the development of resistance in bacteria. Thus, for understanding the process of development of antibiotic resistance, there is a need for an extensive understanding of the genetics involved during the emergence of antibiotic-resistant bacteria.

4. GENETICS OF ANTIBIOTIC RESISTANCE

Resistance of even one bacterium to a particular drug gives rise to the evolution of drug resistance. Resistance genes arising in microbes arise either due to *de novo* mutations or as a means of survival to sustain in an environment where competing bacteria produce antibiotics [59]. These drug-resistant bacteria have become predominant because they survive the course of treatment, and there are very few competitors left. Such antibiotic-resistant bacteria can transmit to another person and eventually the entire host population [60]. The genes that confer ABR are transferred either through horizontal or vertical transfer. The vertical transfer involves the inheritance of DNA from the mother cell to the daughter cells after the mutation of DNA [61, 62]. Horizontal transfer consists of DNA transfer between bacteria of the same or different species in a microenvironment. The horizontal transfer includes three methods- Conjugation, Transformation, and Transduction [63], depicted in Fig. (4).

Normal gut microflora is one of the main reservoirs for disseminating ARGs (antibiotic resistance genes) to pathogenic bacteria. ARGs (macrolide-resistant genes *ermF*, *ermG*, *ermB*, and tetracycline-resistant genes *tetQ* and *tetM*) get transferred between related bacterial species. However, gene transfer between unrelated microorganisms like anaerobic commensals in the gut and Gram-negative anaerobic opportunistic pathogens is unlikely to occur. Some antibiotic-resistant genes are present in the gut of a healthy human and a patient. One such ARG is the β lactamase *cbla*, present in *Bacteroides*. Reports indicate that *E. coli* affecting poultry has the same genealogy as uropathogenic *E. coli* (UPEC), which alludes to the transmission of antibiotic-resistant UPEC through food [64]. Some genes and proteins impart resistance to a particular microorganism like those forming biofilms and promoting resistance or some enzyme encoding genes. For instance, biofilms make it difficult for conventional antibiotics to penetrate the cell, making it less susceptible. *E. coli* is the most frequent causative agent of UTIs (Urinary Tract Infections). The clinical isolates are resistant to several antibiotics, including amoxicillin-clavulanic acid, ampicillin, norfloxacin, ceftriaxone, co-trimoxazole, and cefuroxime. Studies conclude that biofilm containing microbes are resistant to multiple antibiotics. The stress-resistant genes help the microbe develop biofilm in harsh environments like the urinary tract in uropathogenic *E. coli* [65]. The genes responsible for ABR in most *E. coli* strains and other microbes are mentioned in Table 1. Moreover, genes have been determined that are involved in developing antibiotic resistance. Still, understanding the antibiotic resistance mechanism is equally important for developing treatment strategies or approaches.

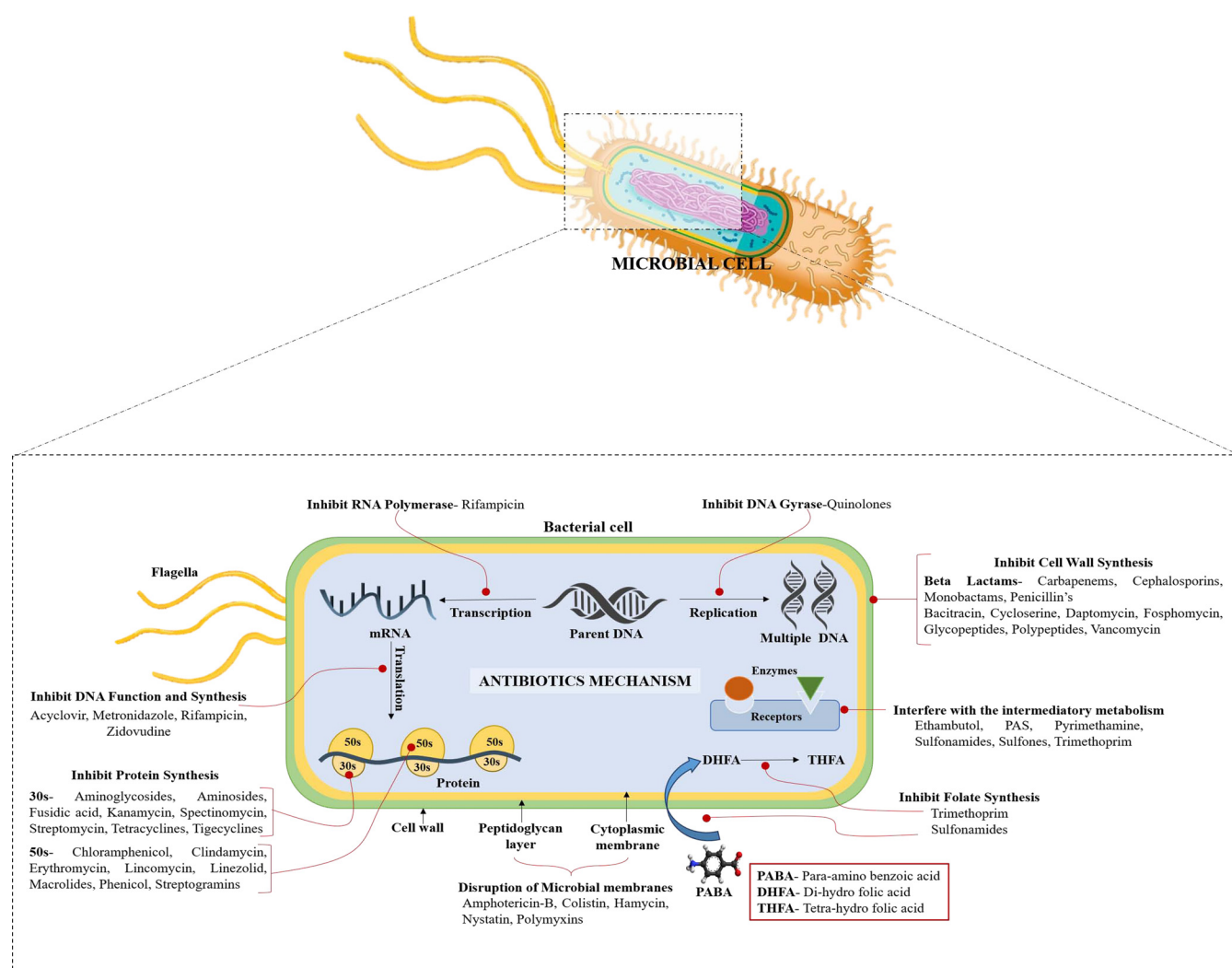


Fig. (3). Mechanisms of action of antibiotics. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

Table 1. The genes involved in antibiotic resistance, their function, and location in different microorganisms.

Organism	Mechanism of Resistance	Gene(s)	Location	Refs.
<i>Escherichia coli</i>	Biofilm formation	<i>ymgB</i> or <i>AriR</i>	Cytoplasm	[65]
<i>Staphylococcus aureus</i>	Beta lactam resistance	<i>mecA</i> , <i>mecR1</i>	<i>Staphylococcal cassette chromosome mec</i>	[66, 67]
<i>Klebsiella pneumonia</i>	Carbapenem resistance	KPC (<i>Klebsiella pneumonia</i> carbapenemase)	Plasmid	[68]
<i>Mycobacterium tuberculosis</i>	Isoniazid resistance	<i>katG</i> and <i>inhA</i>	Plasmid	[69, 70]
<i>Neisseria gonorrhoeae</i>	Tetracycline resistance	<i>tetM</i>	Plasmid	[71]

5. MECHANISMS OF ANTIBIOTIC RESISTANCE

There are two primary levels where bacteria can develop ABR – cellular and community levels. The community level of resistance involves the resistance incurred due to bacterial adaptation to stressful environmental conditions. Bacteria switch from free-living or planktonic forms to a multi-stratified community or bio-film [72]. The cellular level of the mechanism involves resistance through Efflux pumps or changes in cell permeability, inactivation of the antibiotic through enzymatic resistance, and alteration in the

antibiotic target or chemical modification of the antibiotic. Even after the development of community-level resistance, the cellular level of resistance may persist at the single-cell level [73].

5.1. Cellular Level Antibiotic Resistance

5.1.1. Toxin-Antitoxin (TA) System

TA systems are present in most free-living bacteria that provide them the capacity to endure stressful conditions like antibiotic pressure or nutrient deprivation. These systems trigger the evolution

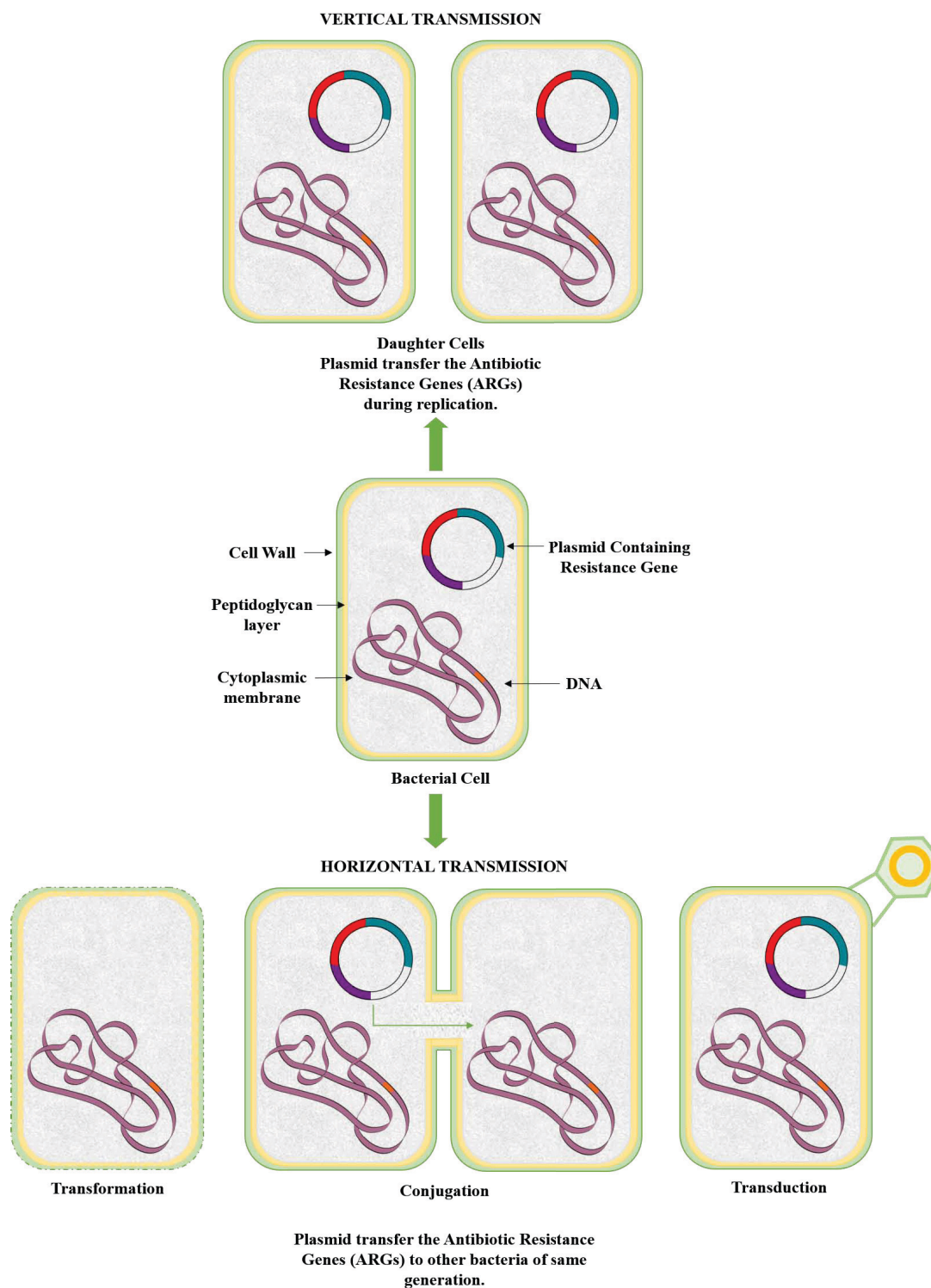


Fig. (4). Transmission of antibiotics resistance genes in bacteria. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

and dissemination of ABR in bacteria in different ways, like an important role in biofilm formation and maintenance of MDR (multi-drug resistant) plasmids [74]. The classification of TA systems as types I-IV is based on the nature of the antitoxin and the mode of toxin inhibition [75]. The toxin molecules or proteins in TA systems are responsible for inhibiting the cell growth through cleavage of DNA or RNA, phosphorylation of proteins,

inhibition of ATP synthesis, and transfer of phosphate group resulting in stasis in a stressful environment [76]. The antitoxin molecules having an antisense RNA strand are the binding partners of toxin molecules on a gene or protein and are unstable compared to the toxin molecules. Hence, to prevent the toxic action of toxin molecules, the antitoxin molecules must be produced continuously [77].

5.1.2. Efflux Pumps

Efflux pumps play a physiological role in removing toxins and metabolites from the cell. However, they also remove the antibiotic molecules from the cell. Efflux pumps are systems in the bacterial genomes that greatly enhance and significantly contribute to resistance to multiple antibiotics. Each efflux pump has three main parts: i) the outer membrane channel, ii) the periplasmic lipoprotein, and iii) the inner membrane transporter [78]. The development of compounds that can overcome the efflux systems in bacteria remains a significant challenge in designing new antibiotics. Efflux pumps lead to the antibiotic release from the cell efflux, causing a decrease in the antibiotic concentration within the cell and increasing the MIC (Minimum inhibitory concentration). An efflux pump can be used for a specific substrate or exit a range of structurally distinct compounds from the cell. The latter leads to the rise of multi-drug resistance [79, 80].

5.1.3. Antibiotic Inactivation

It is a cooperative trait that is spread across the entire bacterial community. Antibiotic inactivation provides intrinsic and acquired resistance and is distributed among some of the most widely used antibiotics, including chloramphenicol, β lactams, and aminoglycosides. The trait of cooperative enzymatic inactivation can be shared across the same or different members of the community [81]. It protects the microbe by decreasing the antibiotic concentration in the environment. For instance, the β -lactamase produced by Enterobacteriaceae inactivates β lactam antibiotics and protects *Salmonella enteric* serovar typhimurium from detrimental concentrations of the antibiotic [82].

5.2. Community Level Antibiotic Resistance

A microorganism's response to the antimicrobials in its environment is often a sum of its community interactions. Factors like the composition of a microbial consortium, its structure, and the ecological and biochemical interactions of the participating species can strengthen the ABR mechanisms. This is typically seen when patients with polymicrobial infections do not respond to clinically validated antibiotic therapies. For example, biofilm formation can increase the MIC for infection [83]. Two types of community interactions are responsible for the failure of action of antibiotics. The first is the development of microbial consortia (or a microbial community), leading to the emergence of antibiotic resistance. These include processes such as biofilm formation and horizontal gene transfer. The second is the host community interactions and mechanisms, which manifest as behavioral factors like misuse of antibiotics, hygiene issues, and host immune responses.

5.2.1. Biofilm Formation

Biofilm is a consortium-level resistance mechanism where surface-attached bacteria are enclosed within an extracellular (self-generated) matrix composed of proteins, polysaccharides, and DNA [84]. Biofilm protects bacteria from many stressors, including antibiotics, the immune system, and disinfectants [74]. The intrinsic resistance offered to conventional antibiotics through the biofilm is because of the bacterial subpopulations present in the deepest layers of the biofilm having lower metabolic activity and proliferation rate [85]. Biofilm represents a niche that promotes the horizontal transfer of genes within the bacterial subpopulations [73]. Hypermutation within the bacterial subpopulation in the biofilm leads to the development of resistance mechanisms like up-regulation of porin proteins and overexpression of efflux pumps [86].

5.2.2. Horizontal Gene Transfer

A horizontal gene transfer is a way to spread the resistance genes across the bacterial community, even crossing the species boundary. The pathogenic bacteria acquire pre-existing resistance

genes from the commensal bacteria to evolve the resistance [87]. Some factors affect the spread of resistance genes through horizontal gene transfer. One, for the transfer to occur from the environmental bacterium to pathogenic bacteria, the two must share the same habitat (at least temporarily to facilitate transfer). Second, the transfer occurs under stress conditions, like in the presence of antibiotics, biocides, etc. Finally, the transfer is more likely to occur between phylogenetically closely related species. The establishment of the transferred genes is stipulated by antibiotic selection [88]. It is unavoidable to prevent the spread of resistance among pathogens since they are phylogenetically related and share similar habitats. It is an astounding fact that the human gut microbiome harbors resistance genes that have yet not been transferred to pathogens. The reasons for this remain unknown. A pathogen that has been extirpated after acquiring resistance from an environmental pathogen and before entering the human again poses no threat. However, a repeated interaction between the resistant bacteria and the human host provides conditions for the selection of resistance. The spread of resistance genes through horizontal gene transfer is accomplished through mobile genetic elements including plasmids, bacteriophages etc [89, 90], illustrated in Fig. (5). Although substantial efforts have been made to understand the mechanism of antibiotic resistance development, continuous efforts still need to be made to determine the emergence of antibiotic resistance.

6. EMERGENCE OF ANTIBIOTIC RESISTANCE

The scarcity of epidemiological data from different world areas makes it difficult to quantify the ramifications of ABR. The development of ABR among microorganisms is a natural phenomenon [91]. Resistant mutations in microorganisms result from *de novo* mutations during replication. However, the emergence of ABR is caused by human behavior resulting in the exploitation of antibiotics [92]. Primarily, resistance evolution leads to the overuse of antibiotics. As early as 1945, it was stated by Alexander Fleming, "the public will demand (drug and) then will begin an era of abuse". Studies reveal that in approximately 30-50% of the cases, the duration of antibiotic therapy is inappropriate [93]. ABR in various fields has been discussed as follows:

6.1. The Emergence of Antibiotic Resistance in the Field of Medicine

To determine the potential causes of ABR, it is essential to understand that there is a sequential pattern followed for a particular drug to reach the patient, and any unwise practice along the way can contribute to ABR [94]. Factors like drug availability, storage, and handling of the drug affect the potency of the drug leading to complications upon consumption by the patient. Many unauthorized services available in developing countries are a major contributing factor to the faceless pandemic of ABR. According to some reports, many antimicrobials in Africa available in pharmacies without an authorized license are controversial or of poor quality. However, such pharmacies are easily accessible to the general public due to certain factors like the absence of consultation fees [95]. Consumption of a drug with degraded quality (a possible consequence of storage or handling errors) results in exposure of microorganisms to sub-therapeutic concentrations of the antibiotic. These degraded drugs contain an altered concentration of the active substance of the antibiotic or might even have the altered form of the active substance. This leads to the emergence of ABR. Data also suggests that in some countries, uncertain diagnosis leads doctors to blindly resort to over-prescribing antibiotics to patients suffering from life-threatening diseases due to trepidation of unsuccessful outcomes [96].

Nosocomial infections due to ABR are a severe threat to health and spread rapidly in hospital settings. Many factors contribute to this emergence, including the prolonged and extensive use of antibiotics in hospitals, the inability of the hospital to prevent the

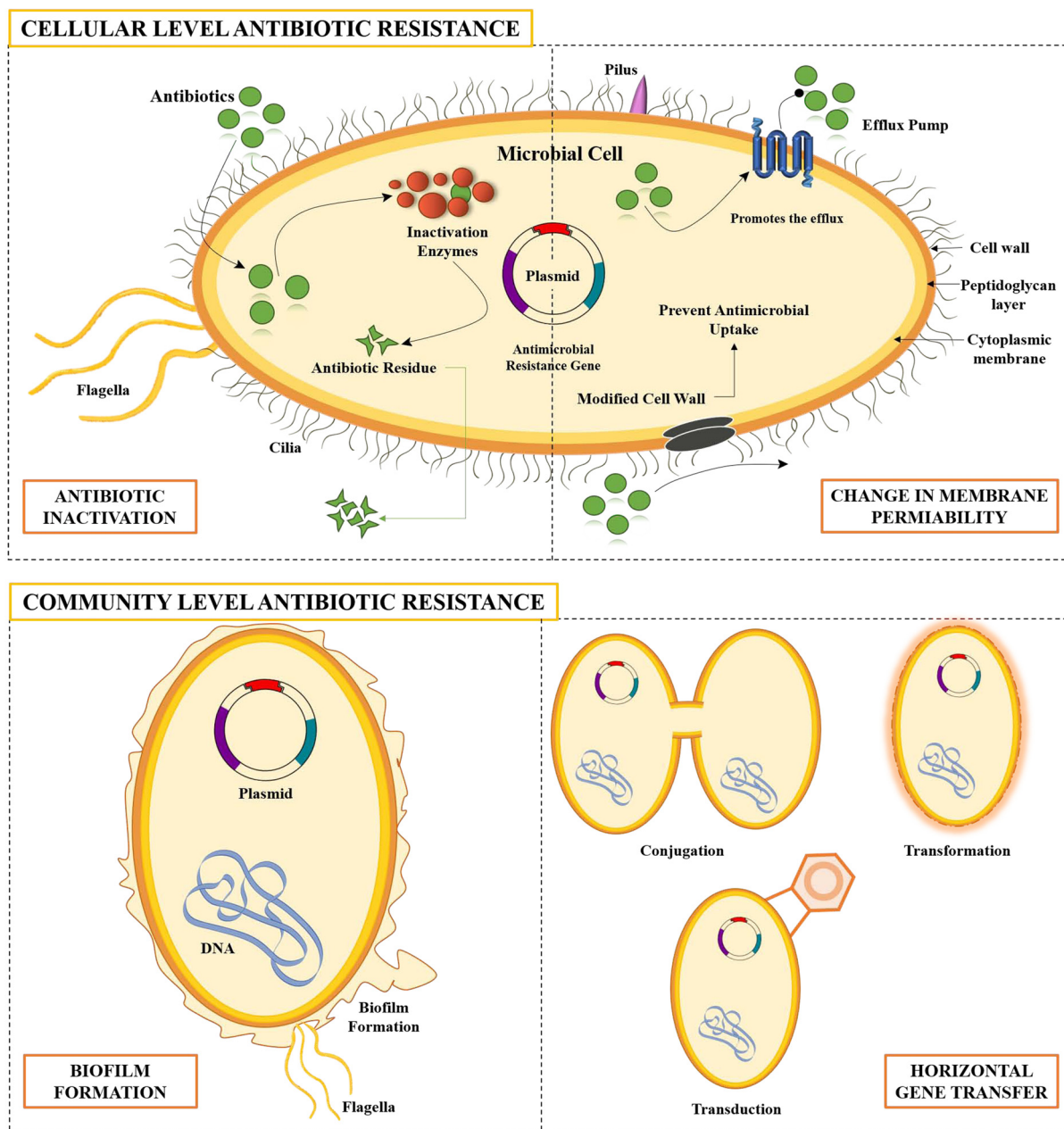


Fig. (5). Mechanism of antibiotics resistance in bacteria. (A higher resolution/colour version of this figure is available in the electronic copy of the article).

spread of infection amongst patients, and the presence of immuno-compromised patients (who have cancer, AIDS, undergoing transplantation surgeries, among others) [97]. Self-medication is another reason for resistance since there is the easy availability of drugs without prescription in most developing countries. Along with self-medication, failure to comply with the recommended course of antibiotics contributes to ABR. Self-medication is usually the result of self-diagnosis, and it often lacks the necessary indications required for the intake of antibiotics. Meta-analysis reports on indiscriminate or non-prescription use of antibiotics reveal that up to 69% of antibiotic supply was without prescriptions in the sub-Saharan African nations, with the most commonly misused antibiotics being amoxicillin and co-trimoxazole [98]. As a result, there is a crucial need for the surveillance of antibiotic consumption across different nations to prepare the scientific community for potential cases of antibiotic resistance.

A study published in The Lancet reported the antibiotic consumption pattern in 101 countries over 19 years. The study said that the highest usage of antibiotics is seen in east Asia, southeast Asia and eastern Europe [99]. A similar study was done in Ethiopia. Out of 780 respondents of the study, 367 had non-prescribed antimicrobial use, which amounts to around 47% [100]. Per a 2018 survey conducted by the WHO, at least six nations reported the regular over-the-counter sale of antibiotics (without prescription). At least eight nations reported the same to occur occasionally. This study was done with 18 European antimicrobial medicine consumption network countries [101]. Such surveillance data can be of immense help for monitoring antibiotic prescriptions, both necessary and unnecessary. However, the research component alone is not sufficient. There is an urgent need to implement stringent regulatory policies on the prescription, supply, and sale of antibiotics. As a start, all antibiotics and antibiotic-based medications (includ-

ing topical formulations) should be made prescription-based globally and should not be available over-the-counter. There is also a dire need for educating pharmacists globally if the scientific community is to prevent a looming albeit quiescent pandemic of antimicrobial resistance.

6.2. The Emergence of Antibiotic Resistance in the Field of Agriculture and Farming

The population's increasing demand for animal protein has instigated intensive livestock farming. Antibiotics are used to prevent animals from diseases, but sublethal concentrations of antibiotics are employed to enhance the growth of animals and improve the efficiency of food conversion. This leads to the pre-disposal of humans with an increased risk of exposure to antibiotic-resistant genes through the food chain [102]. Enhanced usage of antibiotic streptomycin is carried out in agriculture, especially for the prevention of infections caused due to the bacterium *Erwinia amylovora* in apples and pears. The antibiotics are sprayed on the plants, and the agricultural runoff can spread to the neighboring water bodies and other agricultural lands, impacting the environment. The ABR genes can be transferred to clinically significant microorganisms and render the antimicrobials ineffective [103]. There have been instances where antibiotic-resistant bacteria were transferred from animals to humans through the food chain. For example, in 2014, there was an outbreak of multi-drug resistant *Salmonella heidelberg* in the USA. This incident was linked to meat contamination from one supplier due to the extensive use of an antimicrobial. Consumption of infected chicken meat led to this multistate outbreak. Such incidents caused the public to demand for “antibiotic free meat” from the dairy and farm industry [104]. Colistin, a polymyxin type of antibiotic, also referred to as the last resort antibiotic, was reported to be ineffective for the treatment of an infection caused due to gram-negative bacteria. “Apocalypse pig” was the first pig that showed resistance to colistin. This was the last line of treatment against the infection [105].

6.3. The Emergence of Antibiotic Resistance in the Environment

A high concentration of antibiotics in the environment due to the exploitation and indiscriminate use of antibiotics by humans in various fields such as medicine and agriculture is called “antibiotic pollution.” Primarily, humans and animals undergoing antibiotic therapy are responsible for releasing antibiotics into the environment via the contamination of waste streams [106]. This is the prime source of antibiotic pollution because antibiotics are not metabolized completely and are discharged into the environment. India is one of the largest producers of pharmaceutical effluent in the world. Antibiotics with a high affinity for water spread ecotoxicity in the environment faster. The rate at which antibiotics are released into the environment depends upon the dosage and administration of the antibiotic along with animal species [107]. This happens due to the seepage of manure containing antibiotics and antibiotic-resistant genes in the soil. Runoff from farms leads to water contamination and spread antibiotic pollution further. Microorganisms can combat the static or tidal effect by continuous exposure to antibiotics through the alteration of target sites [108]. There is a serious concern about hygiene in today's world.

As a consequence of this, products that claim to have antimicrobial agents are in high demand. However, there is an increased risk associated with environmental health. In September 2016, the FDA in the USA banned the usage of antibacterial soaps containing the antibacterial compound triclosan and 18 other compounds. There were many reasons for the ban imposed on the use of products containing triclosan, but the major threat was the development of antimicrobial resistance. Such compounds target the microbial cells non-specifically; hence it was believed that resistance to them would not occur usually. However, the presence of drug efflux

pumps ruled out the possibility of no resistance. Drug-efflux pumps are the major contributors to antimicrobial resistance [105].

6.4. Other Factors

Certain factors trigger the spread of ABR worldwide. Global health care is highly concerned regarding the emergence and spread of antimicrobial resistance due to the high health threat to the public and undermining of the benefits provided to the society through easy access to antimicrobials [109].

6.4.1. Self-medication

Rapid spreading of ABR is attributed to its misuse (use of an antibiotic without prescription, self-medication, and interrupted dose of antibiotics) due to lack of awareness about ABR. Reports state that ~20-50% of antibiotics used in humans and 40-80% in animals are unnecessary and questionable [110]. In 1994, about 24 kg of vancomycin was actively used for human therapy, and 24,000 kg avoparcin was used as a feed additive for animals. Detection of vancomycin-resistant *enterococci* of animal origin in humans depicts the spread of antibiotic-resistant genes through consumption of contaminated meat [111].

6.4.2. General Misuse of Antibiotics by Physicians

Patients do not adhere to the guidelines of the prescriber leading to inappropriate dosing of the antibiotic [63]. Before the antibiotics were discovered, hospitals were full of patients suffering from infections contracted either due to cuts or scratches. These infections, which were once life-threatening, are now treated easily due to the easy accessibility and administration of antibiotics [112]. One of the essential medical approaches used in the modern medical field to manage diseases like cancer in critically ill patients or surgical procedures and transplantations is antibiotic administration [113]. ICU (Intensive care unit) is the cradle of ABR and is responsible for developing ABR due to high exposure of antibiotics in a heavy density patient population. The unnecessary prescription of antibiotics contributes to the spread of ABR [111].

6.4.3. Poor Quality of Antibiotics

Antibiotics can degrade due to certain factors. Some of them are exposure of antibiotics to temperatures above 25°C by hawkers, new labels on expired drugs, exposure of antibiotics to harsh temperatures during shipment, and dumping of antibiotics without labels. A report states that 8 out of 10 brands of ampicillin in Bangladesh were found to be substandard [114]. Antibiotics with substandard quality, with less or no active ingredients, have been found in countries including Indonesia, Bangladesh, Brazil, Thailand, and Malaysia.

6.4.4. Host Deterioration by Pathogens

Pathogens are well-known producers of toxins (exotoxins and endotoxins), which can cause significant cell damage and death. As a result, inflammation develops. Especially in chronic infections, the inflammation is prolonged, thereby reducing the overall immune response and the synergism between antibiotic therapy and the host immune response. Some well-known examples include the *Neisseria* IgA protease, C3bot (*C. botulinum*), C3lim (*C. limosum*), and *Streptococcal streptolysin* O. The IgA protease from *N. gonorrhoea* (the causative agent of gonorrhea) degrades immunoglobulin A, leading to increased adhesion of the pathogen on epithelial surfaces [115]. The IgA protease also is reported to degrade LAMP-1 *in-vitro*, implying its role in immune-evasion through escaping phagocytosis [116]. C3bot and C3lim belong to the transferase class of enzymes and bears ADP-ribosyltransferase activity. Upon endocytosis in macrophages and monocytes, these toxins derail the rhoA/B/C cytoplasmic signaling through ADP-ribosylation. As a result, these cells lose their phagocytic and migratory functions [117]. Streptolysin O is a pore-forming hemolytic toxin that,

through a cholesterol-dependent mechanism, causes target cell lysis by osmotic shock [118]. Therefore, it targets neutrophils explicitly and causes significant neutropenia. These virulence factors contribute to an uprise in antimicrobial resistance as the susceptible and immunocompromised hosts provide incubation to these pathogens to evolve more immune-evasion maneuvers.

6.4.5. International Travel

Individuals exposed to resistant microbes in one country can spread it to wherever they travel. An increase in intercontinental travel has also increased the spread of ABR. For instance, resistant strains of *Neisseria gonorrhea* originated in the Philippines but now can be found throughout the world [111, 119, 120].

ABR compromises the efficacy of antibiotics for the treatment of infections, resulting in increased health care costs and high mortality rates. Infections resistant to antibiotics have become a significant public health threat and affect people worldwide. Gonorrhea, the second most common infectious disease caused due to *Neisseria gonorrhea*, is cephalosporin-resistant and other antibiotics. *Klebsiella pneumonia* has a resistant mechanism of production of β lactamase that makes it resistant to extended-spectrum β lactamases. *Vibrio cholera*, *Streptococcus pneumonia*, *Mycobacterium leprae* are other microbes that have developed drug resistance posing major health threats [121]. However, ABR is compromising the successful recovery of the patients and spreading a silent pandemic around the globe. By observing the current scenario, a particular focus is required to assess the ramifications of antibiotic resistance and re-emergence of infections by antibiotic-resistant bacteria.

7. RAMIFICATIONS OF ANTIBIOTIC RESISTANCE AND RE-EMERGENCE OF INFECTIONS

The Healthcare field without antibiotics will potentially lose all value and benefits to humankind until now. Even minor cuts will turn into critically serious infections without antibiotics. The present antibiotic era, where resistance to many drugs has already emerged, is challenging future medical advancements. ABR emerged as soon as in the 1940s, only a decade after discovering the first antibiotic, Penicillin [122].

Globally, the developing nations and developed nations are hit by the faceless pandemic of ABR. Prophylactic antibiotics are used for cesarean section, organ transplants, and joint replacement. There is enough evidence stating that cancer chemotherapy is also leading to ABR. The gut microbiota is required to display the optimal potential for immunotherapy and cancer chemotherapy. Hence, intestinal dysbiosis due to these therapies might be due to the absence of such agents [123].

ESBLs (Extended-spectrum β -lactamases) are plasmid-derived enzymes found in gram-negative microbes that inactivate the β lactam ring of antibiotics and render them ineffective. Initially, The ESBLs produced were a narrow spectrum, but microbes produced broad-spectrum lactamases in response to broad-spectrum antibiotics. ESBLs are detected based on their ability to resist third-generation cephalosporins, and the β lactamase inhibitor, clavulanic acid, is detected based upon its ability to block the resistance of ESBL [124]. Carbapenems like imipenem and meropenem are the best treatment options available against ESBLs. Even if cephalosporins, including cefepime and piperacillin-tazobactam, show *in vitro* susceptibility, they must not be administered since the ESBL producing plasmids encode resistance genes to other drugs like aminoglycosides also, and this makes the organism multi-drug resistant [125].

Carbapenemases are β lactamases with the ability to hydrolyze a no. of antibiotics, including penicillins, monobactams, cephalosporins, and carbapenems. Based upon the homology of amino acids, carbapenems can be classified into four major classes, classes

A, B, C, and D. Classes A, C, and D comprise of serine at their active sites, and class B consists of zinc at their active site [126].

Unresponsive strains of *Staphylococcus aureus* were detected in English hospitals. MRSA (Methicillin-resistant *Staphylococcus aureus*) is an intensively studied pathogen and a “mascot” antibiotic-resistant bacteria. It was detected in a Boston hospital and was resistant to the second generation of the antibiotic penicillin. Reports suggest that the frequency of isolates of MRSA increased intensively from close to zero to approximately 70% in Japan and the Republic of Korea, 30% in Belgium, and 40% in the UK and US. Soon, other clinically essential pathogens developed resistance. Infections caused by these microorganisms cannot be treated even by modern-day antibiotics [127]. The drug of last resort, Colistin, which was earlier rejected due to toxic effects, is used to treat these infections. Continuous research for new antibacterial agents during the previous years has led to successful outcomes. However, ABR poses a significant threat to public health in developing nations around the globe where the second and third line of treatment is not available and already available antibiotics are ineffective due to extensive ABR. Developed countries can cope with this faceless pandemic as they can afford the newly developed drugs [128].

MDR (multi-drug resistant) and XDR (extensively drug-resistant) strains of *Mycobacterium tuberculosis* have evolved and become a worldwide health threat. The first line of anti-TB treatment involves using a drug called rifampicin. According to a report, there were 5,00,000 cases of TB worldwide in 2018 that were rifampicin-resistant. The second line of treatment is toxic, poorly tolerated, and administered for ~20 months. The drugs employed in the second line of therapy include Fluoroquinolones, Capreomycin, Kanamycin, and Amikacin. The second line of treatment is only 56% successful compared to the drug-sensitive TB, which is 85% [121, 129].

E. coli is considered an indicator of antimicrobial resistance in gram-negative bacteria. The ABR of *E. coli* is of concern since the resistance mechanism acquired through horizontal transfer have increased considerably in the past 20 years. The transfer of β lactam resistance to nosocomial *E. coli* has made it resistant towards the 3rd generation and 4th generation cephalosporins and carbapenems. The worldwide pandemic strain of *E. coli* type 131 is widely spread among the antimicrobial-resistant community [130].

Vancomycin-resistant *Staphylococcus aureus* (VISA) was first found in 2002, about 44 years after discovering the antibiotic Vancomycin. Multi-drug resistant bacteria are continuously spreading around the globe. The study reveals that an estimated no. of 25,000 patients die in Europe every year due to multi-drug resistant (MDR) infections [131]. Reports suggest that an epidemic of *Shigella diarrhea* in Guatemala claimed the lives of 12,500 people. The causative agent had a plasmid lodged in it that was resistant to four antibiotics [132].

Table 2 listed below depicts various antibiotic-resistant organisms that cause infections and are widespread around the globe.

A recent study using CDDEP showed a variation among the resistance rates of pathogens; the resistance rates of *Acinetobacter baumannii* were higher than those of the other bacteria, and the resistance rates of *Enterococcus faecium* were highest toward broad-spectrum penicillins. *Klebsiella pneumoniae* is included in the priority four pathogens list of the World Health Organization (WHO) as it showed high resistance rates [133].

As far as the Indian scenario related to ABR is concerned and the new challenges conferred by the multi-drug resistant pathogens. India is also combating the old rivals like tuberculosis, malaria, and cholera pathogens, which are resistant to increasing antimicrobials. India has been referred to as the ‘AMR capital.’ According to the ‘scoping report on antimicrobial resistance in India (2017), among the Gram-negative bacteria, more than 70 percent

Table 2. Enlist Main Antibiotics-resistant bacteria with a list of resistant drugs and common infections.

Causative Agent	Resistant Drug(s)	Common Infections
<i>Escherichia coli</i>	• Carbapenem • Cephalosporins • (3rd gen) • Fluoroquinolones	• Food borne infections • Meningitis (in neonates) • Peritonitis (intraabdominal) • Urinary tract infection
<i>Klebsiella pneumoniae</i>	• Carbapenem • Cephalosporins • (3rd gen) • Cotrimoxazole • Fluoroquinolones	• Meningitis (in neonates) • Urinary tract infection
<i>Mycobacterium tuberculosis</i>	• Fluoroquinolone • Isoniazid • Rifampicin • Streptomycin	• Tuberculosis
<i>Neisseria gonorrhoeae</i>	• Cephalosporins (3rd gen) • Fluoroquinolones • Penicillin • Tetracycline	• Acute reproductive tract infection • Eye infection (in new born) • Infection in pharynx and rectum (in new born)
<i>Shigella</i>	• Cotrimoxazole • Fluoroquinolones	• Gastroenteritis • Neurologic disorders
<i>Staphylococcus aureus</i>	• Methicillin	• Food poisoning • Postoperative wound infections • Toxic shock syndrome
<i>Streptococcus pneumoniae</i>	• Penicillin	• Acute otitis media • Community-acquired pneumonia • Meningitis

of isolates of *Escherichia coli*, *Klebsiella pneumoniae*, and *Acinetobacter baumannii*, and nearly half of all *Pseudomonas aeruginosa* were resistant to fluoroquinolones and third-generation cephalosporins. ~71% cases of *Acinetobacter baumannii* are resistant to carbapenems. This has led to increased use of the last-resort antibiotic colistin. Resistance to colistin was rare in India (< 1 percent), but this situation aggravated the rate of colistin resistance. *Salmonella typhi* and *Shigella* species expressed opposition at 28 and 82 percent rates, respectively, for ciprofloxacin, 0.6 and 12 percent for ceftriaxone, and 2.3 and 80 percent for cotrimoxazole. Resistance rates to tetracycline for *Vibrio cholerae* varied from 17 to 75 percent in different parts of the country [134]. The complication and re-emergence of antibiotic resistance have significantly gained interest and demand the development of preventive and control measures.

8. PREVENTION AND CONTROL MEASURES

Scientists state that ABR is inevitable since it results from a natural adaptation. However, there are ways to slow down the evolution process to prevent the rapid spreading of ABR. As an initiative to control the rapid spread of ABR, WHO launched the Global AMR Surveillance System (GLASS) in 2015. This was a standardized approach where 42 countries agreed to share global data on AMR and drive actions on various levels. This system represents a shift from the surveillance approach based only on laboratory data to a system including epidemiological, population, and clinical based data [63].

The Centre for Disease Control (CDC) states four core actions for preventing AMR. They are mentioned as follows:

- 1) Prevention of infection and spread of Resistance- the probability of using an antibiotic depends upon the severity of the infection. The absence of infection does not require the use of an antibiotic. Infections can be prevented by following basic hygiene measures like proper handwashing before and after having food and food preparation, following the correct immunization schedule, and using antibiotics only if prescribed.
- 2) Tracking- Keeping a proper track of the data can help scientists gather new strategies for combating the spread of infection.
- 3) Improvement of Antibiotic Prescription- The spread of infections can be significantly reduced by 'Antibiotic Stewardship'. This states that the antibiotic should be used appropriately and safely. They must be used only when required and administered correctly and in dosage.
- 4) Development of new drugs and diagnostic tests- Since ABR is an outcome of evolution, it cannot be stopped, but the speed can be slowed down. The development of new drugs can help tackle antibiotic-resistant infections (www.cdc.gov).

The WHO described some international measures to tackle AMR:

- 1) Increase the number of collaborations between governmental and non-governmental agencies and international committees.
- 2) Enhancement in the provision of incentives for research and development of new drugs.
- 3) Development of new programs to combat AMR.
- 4) Surveillance of the use of antimicrobials through new networks.

The effective control strategies have been complicated due to the spread of resistance in different environmental niches. This requires the urgent attention of the major agencies [135]. Other than that, there is a need for exploration and development of therapeutic strategies needed to fight against the problem of antibiotic resistance.

9. THERAPEUTIC STRATEGIES TO COMBAT ANTIBIOTIC RESISTANCE

New antibiotic approaches are required to escalate the no. of multi-drug resistant bacteria. The development pipeline consists of many new antibiotic molecules and those in preclinical development. There is still a requirement for the expansion of more clinical options. The adversity of MDR cannot be solved by biologically active molecules alone. It requires new strategies like controlled delivery and antibiotic stewardship.

9.1. Delivery Strategies and Duration of Antibiotic Treatment

The conventional antibiotic administration involves systemic delivery, which is, unfortunately, more damaging than beneficial. Systemic administration of antibiotics does not penetrate to the postoperative or ischemic tissues, which poses the problem of systemic toxicity and poor patient compliance. The subtherapeutic concentrations of antibiotics contribute to the development of ABR further aggravates the infectious complications. The efficiency of antibiotics increases when administered locally [136]. Antibiotics like gentamicin, tobramycin, amoxicillin, vancomycin, cephamandol, and cephalothin have been incorporated into the local release system. As early as the 1940s, efforts were made in dentistry and orthopedics to endow implantable materials with biological activity. However, it soon became evident that the efficacy of these materials was dependent upon the pharmacokinetics of the system.

The local release of an antibiotic must be high, initially, above the MIC (Minimum Inhibitory Concentration) of the infection-causing bacteria and below the toxic threshold level of surrounding host cells, followed by the sustained release to combat the latent bacteria. The duration and sustained level of antibiotic release depend upon factors like the type and mechanism of antibiotic and host tissue matrix. However, the local administration of some antibiotics has caused the initial burst followed by subsequent subtherapeutic activities promoting the ABR [136]. The duration of antibiotic treatment is vital to combat resistance. Procalcitonin (PCT) is an essential biomarker as it distinguishes between bacterial and nonbacterial inflammations. This prevents the unnecessary prescription of antibiotics. The concentration of serum procalcitonin correlates with the severity of infection. Hence, bacterial infections are characterized by high levels of PCT. The level of PCT is below 0.1 ng/mL in patients with no infection. PCT reaches its highest concentration after 14-25 hrs of bacterial stimulus and can be easily detected after 3-4 hrs of bacterial stimulus. Meta-analysis studies showed that PCT is the most promising biomarker and is safe and effective, resulting in reduced antibiotic initiation and short antibiotic duration [137, 138].

9.2. Monotherapy and Combination Therapy

Monotherapy includes the treatment of the patient with only one antibiotic, whereas combination therapy involves the treatment with the simultaneous use of two or more antibiotics. The superior-

ity of monotherapy and combination therapy remains controversial [139]. A study to determine the efficacy of monotherapy and combination therapy for multidrug-resistant gram-negative infections concluded that combination therapy with two antimicrobials must be administered in patients suffering from bloodstream infections carbapenemase-producing Enterobacteriaceae. In monotherapy, there are high chances of prescription of inappropriate empirical antibiotics, and it involves a long duration to start the antibiotic treatment after susceptibility testing [140]. Patients suffering from polymicrobial infections due to resistant and susceptible strains are also prescribed combination therapy as it is favored due to the chance of selection bias [141]. Even though conventional therapeutic strategies are available to contend antibiotic resistance, alternative approaches other than antibiotics are being explored to curb the re-emergence of antibiotic resistance challenges.

10. PREDICTION OF AMR

With the development of new antibiotics against the evolving strains of pathogenic microorganisms, the crisis to develop antibiotics rises as the pace of evolution is more heterogeneous than drug development. A new ML-based prediction strategy has shown significant benefit against AMR [142]. ML-based strategies aid in developing models concerning changing phenotype (varying clinical characteristics) [143] and using sequencing data for identifying changes in the genomic region between the wild strains and evolving strains [144, 145]. Additionally, AI-based approaches may provide insight into the different combinations of antibiotics against the evolving AMR strains and the development of new antibiotics and predict the drug target area. Since such technologies are on the verge of development and new features and algorithms are coming out every day, highlighting both ML and AI may help researchers dig into more complex AMR problems. The only constraint in ML and AI-based research is databases and data redundancy between different databases, affecting accuracy. Still, no doubt, such tools will be a boon for human wellness in the upcoming time.

CONCLUSION

Given the current scenario, the global burden of ABR, and infectious diseases, it has become apparent that there are multiple reasons behind the present situation. Distinct political and socioeconomic factors have led to ABR in developing and developed countries. Considering the different covariates that have implications on the evolution of microorganisms with drug resistance, it is pertinent to understand the pattern of evolution using predictive models. The primary reason leading to ABR involve clinical misuse, easy and unregulated availability of antibiotics, sub-standard quality of available antibiotics, and insufficient surveillance of antibiotic sale and prescriptions. Virulence factors of the pathogens and host immune factors contribute to the development of community-level antibiotic resistance. Research on the mechanisms of antibiotic resistance has revealed the key genes and proteins involved in AMR development, necessitating the development of drugs against the evolving strains. Additionally, the regulatory system needs to improve according to the specific challenges imposed in the particular country. Moreover, there is a dire need to bring awareness among people, especially the nursing staff and pharmacists, by educating them regarding the effective value of antibiotics and preventing the overexploitation of antibiotics so that the rate of resistance development could be lower globally. Therefore, to combat ABR, there is a need to take two crucial initiatives, *i.e.*, promoting research on novel antibiotics and improving the regulatory framework to regulate the usage of antibiotics worldwide. Antibiotic resistance is a pandemic waiting to cause potentially irreplaceable damage to human life. It is imperative that the required policy framework with grassroot implementation is ensured as the burden on healthcare system and the economic burden on people could be potentially irreversible.

LIST OF ABBREVIATIONS

ABR	=	Antibiotic Resistance
ARGs	=	Antibiotic Resistance Genes
CDC	=	Centre for Disease Control
ESBLs	=	Extended-spectrum β -lactamases
ICU	=	Intensive Care Unit
MDR	=	Multi-drug Resistant
MIC	=	Minimum Inhibitory Concentration
MRSA	=	Methicillin-resistant <i>Staphylococcus aureus</i>
PBPs	=	Penicillin-binding Proteins
UTIs	=	Urinary Tract Infections
WHO	=	World Health Organization

CONSENT FOR PUBLICATION

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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